

SKILL 8

Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid, result;

    pid = fork();

    if (pid == 0) {
        printf("Child process is running\n");
        sleep(2);
        printf("Child process completed\n");
    }
    else if (pid > 0) {
        printf("Parent is waiting for child\n");

        result = waitpid(pid, NULL, 0);

        if (result == pid)
            printf("Parent: Child process finished\n");
        }
        else{
            printf("Fork failed\n");
        }
        return 0;
    }
```

Output:

```
shashank@LAPTOP-24KPR0R8:~$ gcc wait[id.c -o wait[id
shashank@LAPTOP-24KPR0R8:~$ ./wait[id
Parent is waiting for child
Child process is running
Child process completed
Parent: Child process finished
```